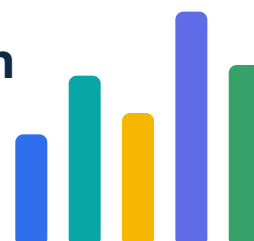


Power BI Portfolio Documentation

Retail Sales & Target Performance Dashboard

Project type: Business Intelligence Dashboard

Tools: Power BI, Excel, Power Query, DAX, Data Modeling



Prepared for portfolio presentation

Retail Sales & Target Performance Dashboard using Power BI

A clean, recruiter-friendly documentation file explaining the business problem, data model, dashboard pages, measures, insights, and skills demonstrated in the project.

Prepared for	Portfolio presentation / LinkedIn / Data Analyst applications
Dashboard pages	Executive Overview, Sales Trend Analysis, Channel & Category Analysis, Target Analysis
Project focus	Sales performance, profitability, channels, campaigns, and target tracking

1. Project Overview

This project is an interactive Power BI dashboard designed to analyze retail sales performance, profitability, product categories, sales channels, marketing campaigns, and revenue targets. The dashboard converts raw transactional data into a clear business intelligence report that supports performance monitoring and faster decision-making.

The report is built as a portfolio-ready project, demonstrating how a data analyst can move from raw Excel data to a structured semantic model, DAX measures, interactive visuals, and business-focused insights.



Business Problem

- Which product categories and sales channels contribute most to revenue?
- How does revenue change over time?
- Are actual sales meeting planned revenue targets?
- Which campaigns generate stronger sales performance?
- How can executives quickly monitor KPIs such as revenue, profit, orders, and achievement percentage?

2. Tools and Workflow

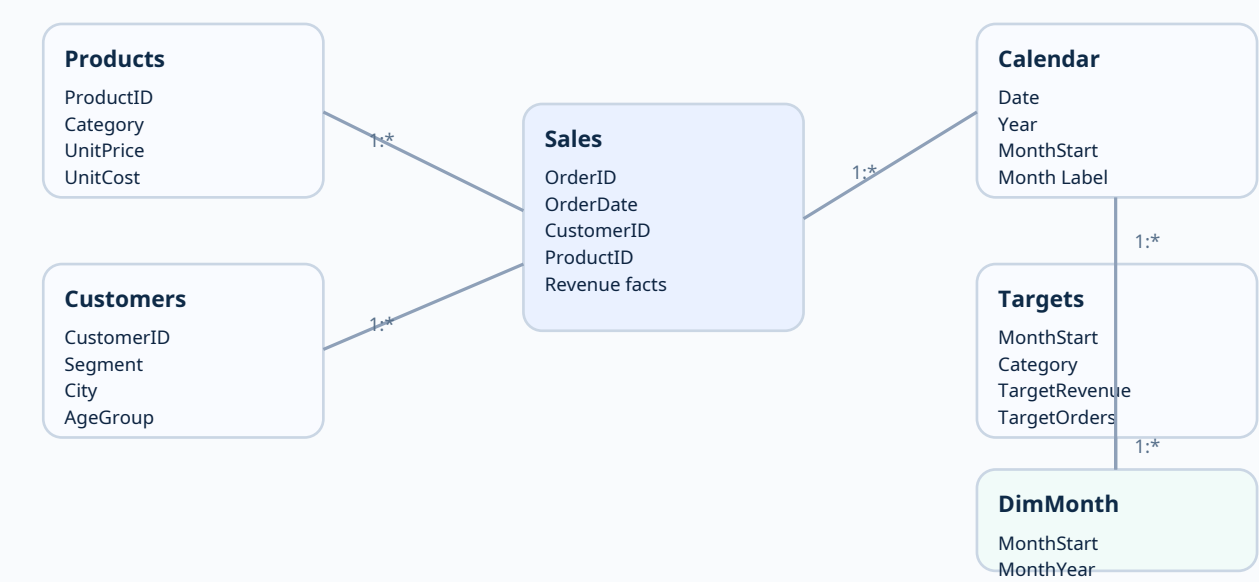
Tool / Technique	Purpose in the Project
Excel	Source file containing structured retail data tables.
Power Query	Data loading, type correction, and preparation before modeling.
Power BI Data Model	Relationships between fact and dimension tables using a clean model structure.
DAX	Measures for revenue, cost, profit, margin, orders, targets, and achievement.
Power BI Visuals	Interactive pages, KPI cards, slicers, trend charts, bar charts, and donut charts.

Workflow Summary

- Imported only the required tables from the Excel workbook.
- Removed unnecessary or duplicated tables to keep the model clean.
- Created a calendar table and a monthly dimension table for target analysis.
- Built relationships between sales, products, customers, calendar, targets, and month dimensions.
- Created a dedicated measures table to organize all DAX calculations.
- Designed four dashboard pages with consistent layout, slicers, cards, and visuals.

3. Data Model and Relationships

The model uses a fact-dimension structure. The Sales table works as the main transaction table, while Products, Customers, Calendar, and DimMonth support filtering, slicing, and time-based analysis. Targets are connected through DimMonth because targets are stored at a monthly level.



Relationship	Cardinality	Purpose
Products[ProductID] -> Sales[ProductID]	1 to many	Analyze sales by product, category, price, and cost.
Customers[CustomerID] -> Sales[CustomerID]	1 to many	Analyze revenue and customers by segment, city, country, and age group.
Calendar[Date] -> Sales[OrderDate]	1 to many	Support revenue and order analysis over time.
DimMonth[MonthStart] -> Calendar[MonthStart]	1 to many	Connect month-level analysis to the daily calendar table.
DimMonth[MonthStart] -> Targets[MonthStart]	1 to many	Compare actual performance with monthly targets.

4. Core DAX Measures

Measure	Business Meaning
Total Revenue	Total sales revenue calculated from order quantity and product unit price.
Total Cost	Total product cost calculated from order quantity and unit cost.
Gross Profit	Revenue remaining after subtracting total cost.
Gross Margin %	Profitability ratio: gross profit divided by total revenue.
Total Orders	Distinct count of customer orders.
Total Quantity	Total number of sold units.
Average Order Value	Average revenue generated per order.
Customer Count	Distinct count of customers.
Revenue per Customer	Average revenue generated per customer.
Target Revenue	Planned revenue target from the targets table.
Revenue Gap	Difference between actual revenue and target revenue.
Target Achievement %	Actual revenue divided by target revenue.

Example DAX logic:

```
Total Revenue = SUMX(Sales, Sales[Quantity] * RELATED(Products[UnitPrice]))
Gross Profit = [Total Revenue] - [Total Cost]
Target Achievement % = DIVIDE([Total Revenue], [Target Revenue])
```

5. Dashboard Pages

Page	Main Purpose	Key Visuals
Executive Overview	Provides a high-level summary for quick business review.	KPI cards, revenue by category, revenue by channel, year slicer.
Sales Trend Analysis	Explores how revenue and orders change over time.	Monthly revenue trend, monthly orders trend, revenue/order KPI cards.
Channel & Category Analysis	Compares performance across channels, categories, and campaigns.	Donut chart, bar chart, campaign column chart, year/category/channel slicers.
Target Analysis	Tracks actual performance against planned revenue targets.	Target cards, revenue gap, actual vs target chart, achievement trend.

Portfolio Page Structure

For a professional portfolio, present the project using screenshots or a PDF export from Power BI, followed by a concise explanation of the business problem, model, DAX measures, and insights.

- Add one screenshot for each dashboard page.
- Include a short paragraph explaining what the page answers.
- Mention the key technical skills demonstrated.
- Add the Power BI file or PDF export when possible.

6. Key Insights

- Total revenue reached approximately \$1.37M across the analyzed period.
- Gross profit reached approximately \$611.52K, with a gross margin of about 44.77%.
- Revenue varies across product categories, helping identify stronger and weaker categories.
- Sales channels contribute differently to total revenue, supporting channel performance decisions.
- Campaign analysis helps compare which promotional activities generated stronger revenue.
- Target analysis shows a clear revenue gap, which is useful for planning and performance improvement.

Skills Demonstrated

Skill Area	Evidence in the Project
Data Modeling	Clean relationships between sales, products, customers, calendar, targets, and DimMonth.
DAX	Measures for revenue, cost, profit, margin, orders, target achievement, and revenue gap.
Business Analysis	Dashboard answers performance, trend, channel, campaign, and target questions.
Visualization Design	KPI cards, slicers, trend lines, bar charts, donut charts, and comparison visuals.
Portfolio Storytelling	Project documentation explains the why, how, and business value of the dashboard.

7. Portfolio-Ready Project Description

Retail Sales & Target Performance Dashboard using Power BI

Built an interactive Power BI dashboard to analyze retail sales, profitability, customer activity, product categories, sales channels, campaigns, and revenue targets. The project includes a structured data model, calendar table, monthly target dimension, and DAX measures for revenue, cost, profit, margin, orders, average order value, customer count, target revenue, revenue gap, and target achievement percentage.

The dashboard contains four report pages: Executive Overview, Sales Trend Analysis, Channel & Category Analysis, and Target Analysis. It demonstrates end-to-end business intelligence skills, including data preparation, relationship modeling, KPI design, interactive reporting, and business insight communication.

Suggested LinkedIn Caption

I completed a Power BI portfolio project focused on retail sales and target performance analysis. The dashboard includes executive KPIs, monthly revenue and order trends, channel and category performance, campaign analysis, and actual vs target tracking. This project helped me strengthen my skills in data modeling, DAX, Power Query, KPI design, and business intelligence storytelling.

Recommended Portfolio Attachments
Power BI file (.pbix)
Exported PDF of the dashboard pages
This documentation PDF
Four dashboard screenshots